Color Fusion

EARLY FDM COLOR MIXING EXPERIMENTS

I've Been Experimenting with FDM Color Mixing!

Hey everyone! I've been experimenting with a way to create colorful prints on standard FDM 3D printers by stacking thin, semi-translucent layers of regular PLA. Here is my latest 8-color test piece fresh off the bed!

Original Digital Artwork — 2D Source (1:1)

Actual 3D Print (180 mm × 180 mm) — P1S + 2 AMS PRINT

The Core Idea: True Physical Layer Blending

Instead of swapping hundreds of single-color spools or hand-painting layer heights, Color Fusion calculates how light passes through thin thermoplastic sub-layers (0.08 mm). By automatically sequencing and stacking ordinary PLA, it blends continuous skies, terracotta rooftops, and sea water reflections right on the build plate!

Real Print Specs

Printer: Bambu Lab P1S with 2 × AMS (8 Slots)
Print Time: 4.5 Hours @ 0.08 mm layer height (180 × 180 mm)
Infill: 100% Solid Rectilinear • 0.4 mm standard nozzle

- eSun Pink (Base)
- Jadeway Matte Yellow
- Bambu Mistletoe Green
- Bambu Matte Beige
- eSun Red
- Bambu Blue Gray
- Jadeway Yellow
- Bambu Jade White

How Does It Work?

You don't need to manually paint pixels or guess layer heights. Here is the automated workflow:

01
Load Any PLA
Pick regular PLA spools from your rack (Bambu, eSun, Jadeway, etc.).

02
Scan SpoolCard
Print a 50×50 mm card to measure light transmission at 10 depth steps.

03
Auto-Compute
Color Fusion searches recipe stacks to recreate the picture's colors.

04
Slice & Print
Export standard 3MF directly into Bambu Studio or OrcaSlicer.

Growing Filament Calibration Swatches (SpoolCard v3.1 — 50 mm × 50 mm)

Bambu Green

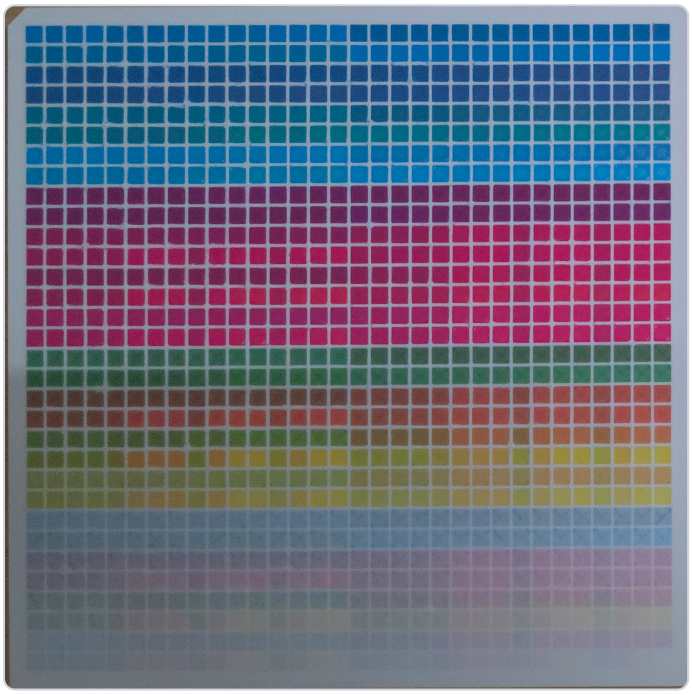
Mistletoe Green

Blue Gray

eSun Red

Sunflower Yellow

Cocoa Brown



1,024 Blended Colors Benchmark: Real 32×32 grid printed in 4 spools, verifying the software's multi-layer light predictions across the entire color spectrum.

Color Fusion Studio Workbench: The web interface where I assign AMS slots, preview sub-layer stacks, and export production-ready 3MFs.

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BEHIND THE SCENES

Early Prototype → Latest Experiment: Real Iteration & Tuning

From Stepped Pixelated Blocks to Smooth Gradients

Getting plastic layers to blend naturally took a lot of trial, error, and algorithm tuning. Here is a direct comparison showing the progress between an early test print and the latest result:

Early Prototype (Oct 15) — STEPPED & BLOCKY

Early Result: Harsh pixel boundaries, coarse color bands across the sky, and oversaturated teal water.

Latest Experiment (Oct 20) — SMOOTH 8-COLOR

Current Result: Continuous sunset sky gradient, natural slate-blue water reflections, and sharp rooftops on P1S (4.5h).

Known Things I'm Still Dialing In:

- **Shadow Tones:** Deep shadows currently print slightly darker than simulated due to heavy pigment stacking.
- **Nozzle Purge Volumes:** AMS flush calibration is crucial; insufficient flushing can cause faint color bleeding into white base layers.
- **Brand Pigment Densities:** Different manufacturer formulas have unique opacities, which is why scanning quick SpoolCards is essential.

Want to see more or share feedback?

- What do you think of this approach to FDM color mixing?
- **I'm considering sharing some test 3MF files** down the road if community members with AMS setups are interested in test-printing!
- What image styles, landscape photos, or filament brands would you love to see tested next?